

Breast Cancer Classification using Artificial Neural Networks

B.Tech Artificial Intelligence – Mini Project Report

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Abstract

This report presents a beginner-friendly artificial neural network (ANN) for binary classification of breast tumors as malignant or benign using the Breast Cancer Wisconsin dataset. Features are standardized, the model is trained with the Adam optimizer, and performance is evaluated using accuracy, loss curves, a confusion matrix, and a classification report. A simple predictive system returns the diagnosis with class probabilities and confidence.

Contents

1	Introduction	3
2	Dataset	3
3	Methodology	4
3.1	Preprocessing	4
3.2	Model Architecture	4
4	Training	5
5	Results	7
5.1	Test Performance	7
5.2	Confusion Matrix and Classification Report	8
5.3	Predictive System	9
6	Project Workflow	11

7 Conclusion

12

1 Introduction

Breast cancer diagnosis from clinical measurements is a classic machine-learning problem. In this project we build a compact feed-forward neural network that maps 30 numeric tumor features to one of two classes: **0 = Malignant** and **1 = Benign**. The implementation uses TensorFlow/Keras and scikit-learn, and is designed to be easy for beginners to follow end-to-end.

2 Dataset

We use the Breast Cancer Wisconsin (Diagnostic) dataset available in scikit-learn [1]. It contains 569 samples and 30 real-valued features describing cell nuclei characteristics derived from digitized images of fine needle aspirates.

Dataset Information

```
Breast Cancer Wisconsin Dataset Information
=====
Total samples           : 569
Number of features      : 30
Feature type            : Numeric tumor measurements
Classes                 : 0 = Malignant, 1 = Benign
Class counts (full)     : Malignant=212, Benign=357

Training samples        : 455
Test samples            : 114
Train labels shape      : (455,)
Test labels shape       : (114,)

Preprocessing           : StandardScaler (fit on train only)
Train/Test split        : 80% / 20% (stratified)
TensorFlow version      : 2.21.0
```

Figure 1: Dataset information summary generated by the build pipeline.

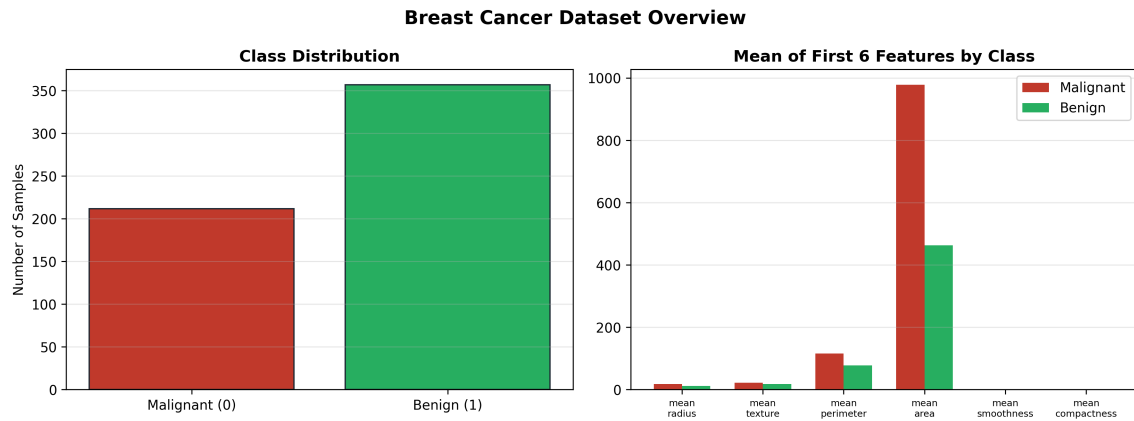


Figure 2: Class distribution and mean feature comparison for selected attributes.

3 Methodology

3.1 Preprocessing

Data are split into training (80%) and test (20%) sets with stratification to preserve class balance. Features are scaled with `StandardScaler` fitted only on the training set to avoid data leakage.

3.2 Model Architecture

The network is a sequential ANN:

- Input layer: 30 features
- Hidden layer: 20 neurons with ReLU activation
- Output layer: 2 neurons with Softmax activation

The model is compiled with the Adam optimizer and sparse categorical cross-entropy loss.

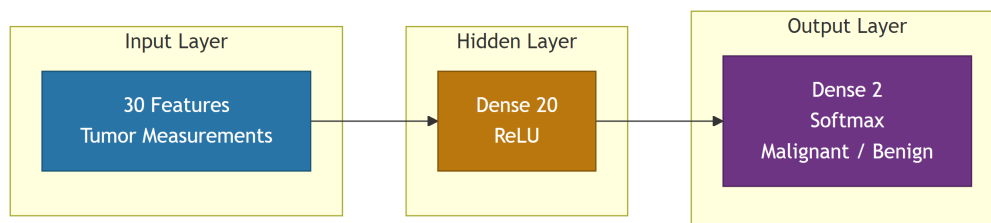


Figure 3: ANN architecture used for breast cancer classification.

Model Summary

Model: "breast_cancer_ann"

Layer (type)	Output Shape	Param #
hidden_layer (Dense)	(None, 20)	620
output_layer (Dense)	(None, 2)	42

Total params: 662 (2.59 KB)
Trainable params: 662 (2.59 KB)
Non-trainable params: 0 (0.00 B)

Figure 4: Keras model summary showing layer shapes and parameter counts.

4 Training

The model is trained for 10 epochs with a 10% validation split from the training data. Accuracy and loss are monitored on both training and validation sets.

Training Progress (10 Epochs)

Epoch	Train Acc	Val Acc	Train Loss	Val Loss
1	52.81%	65.22%	0.7284	0.5902
2	78.97%	86.96%	0.4975	0.4157
3	89.49%	91.30%	0.3653	0.3295
4	92.18%	93.48%	0.2903	0.2803
5	93.64%	93.48%	0.2438	0.2480
6	93.89%	93.48%	0.2121	0.2244
7	94.62%	93.48%	0.1887	0.2055
8	95.11%	93.48%	0.1703	0.1897
9	95.60%	93.48%	0.1555	0.1760
10	96.33%	95.65%	0.1431	0.1639

Final Training Accuracy : 96.33%
Final Validation Accuracy : 95.65%

Figure 5: Per-epoch training and validation metrics.

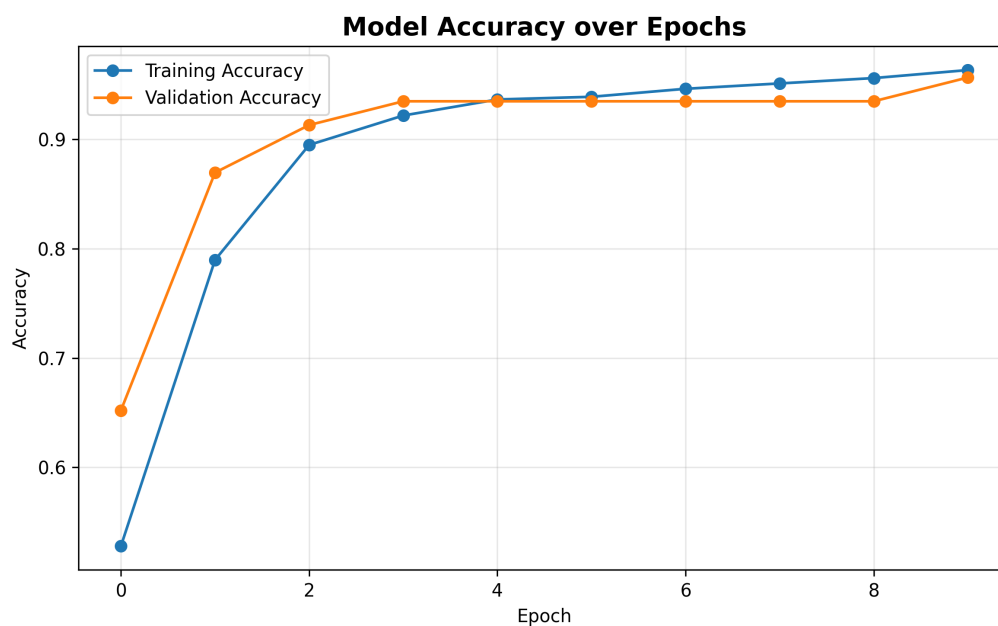


Figure 6: Training and validation accuracy over epochs.

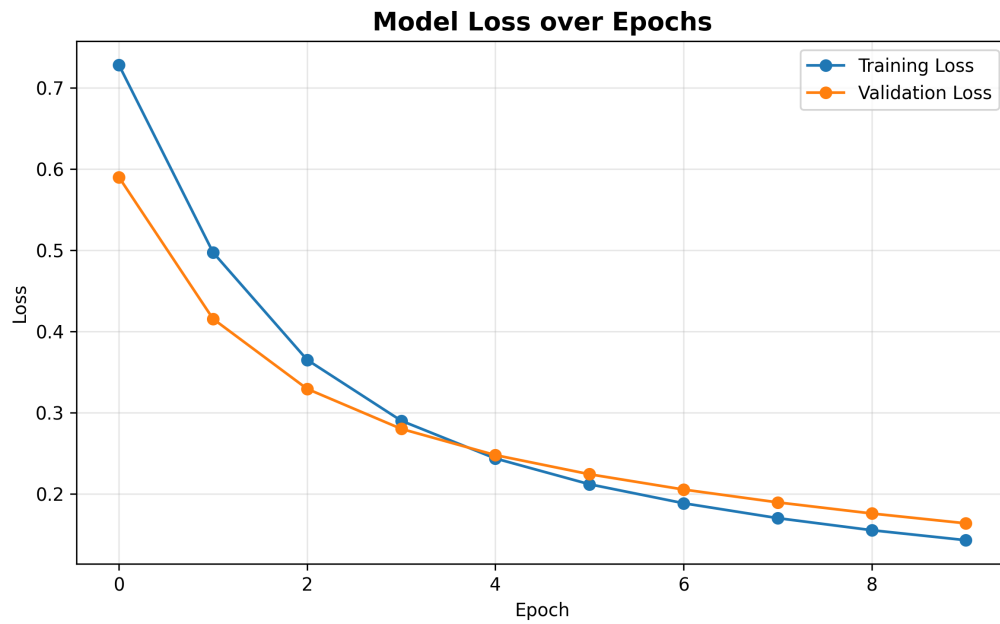


Figure 7: Training and validation loss over epochs.

5 Results

5.1 Test Performance

Test Accuracy

Test Set Evaluation

=====

Test Loss : 0.1077

Test Accuracy : 97.37%

Correct predictions : 111 / 114

Misclassifications : 3

Figure 8: Final evaluation on the held-out test set.

5.2 Confusion Matrix and Classification Report

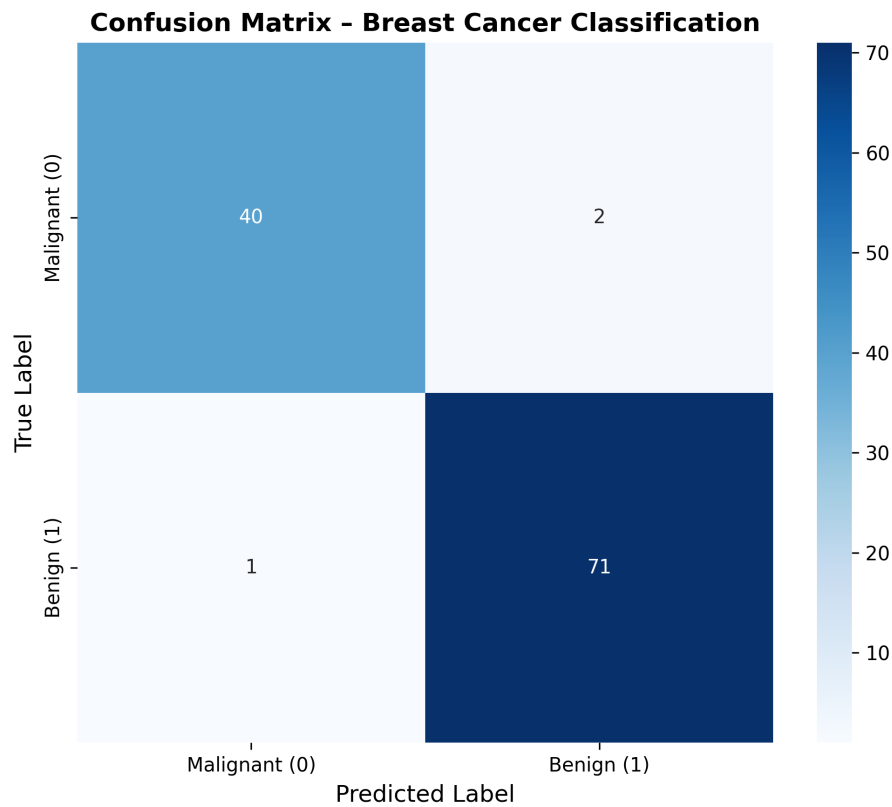


Figure 9: Confusion matrix for malignant vs benign predictions.

Classification Report

	precision	recall	f1-score	support
Malignant (0)	0.9756	0.9524	0.9639	42
Benign (1)	0.9726	0.9861	0.9793	72
accuracy			0.9737	114
macro avg	0.9741	0.9692	0.9716	114
weighted avg	0.9737	0.9737	0.9736	114

Figure 10: Precision, recall, and F1-score for each class.

5.3 Predictive System

A single patient sample is standardized with the fitted scaler and passed to the trained model. The system prints the predicted diagnosis, confidence, and class probabilities.

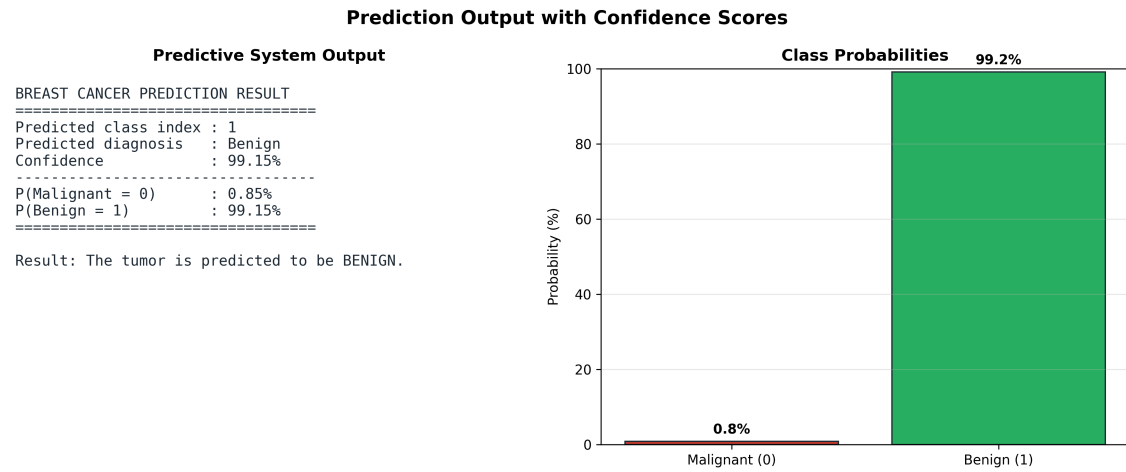
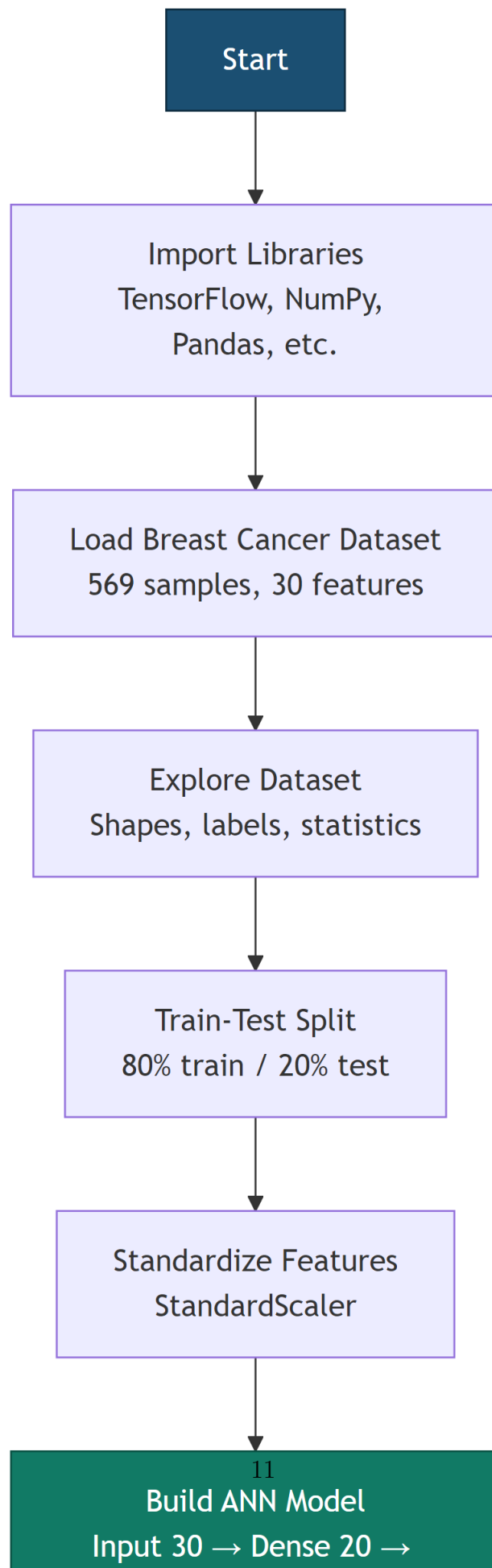


Figure 11: Example prediction output with confidence scores.

6 Project Workflow



7 Conclusion

A simple ANN achieves strong test accuracy on the Breast Cancer Wisconsin dataset and provides interpretable probability outputs for each diagnosis. This pipeline demonstrates data preprocessing, neural network training, evaluation, and a practical predictive interface suitable for academic mini-project demonstration.

Disclaimer

This work is for educational purposes only and is **not** a medical diagnostic tool.

References

- [1] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and E. Duchesnay. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.